Perception First: A Frontier Native-Video Model with Self-Consistency for Implicit Video Question Answering

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Abstract

We describe our submission to the VRR Challenge @ CVPR 2026, built on the ImplicitQA / VRR-QA *benchmark [1]: multiple-choice video question answering in which answers are deliberately* not *observable in any single frame and must be inferred from spatial layout, motion, depth, viewpoint, causality, and social context across discontinuous frames of creative video. We conduct a systematic, training-free study spanning open-source Video-LMMs (Qwen2.5-VL [2], Qwen3-VL [3], InternVL3, Gemma-3, and the RL-tuned video reasoners Video-R1 [4] and VideoChat-R1.5 [5]) and a battery of inference-time strategies (chain-of-thought, question decomposition, describe-then-reason cascades, audio transcripts, spatial state prompting, self-consistency [6], multi-model ensembling, and category routing). Our central finding is that this benchmark is* perception-bound rather than reasoning-bound*: reasoning-side augmentations are neutral-to-harmful, whereas base-model perceptual capability and lightweight test-time denoising are the only reliable levers. The open-source ceiling we reach is* 58.5% *accuracy (Qwen3-VL-32B-AWQ with self-consistency). Our final system—a current* native-video *frontier model, Gemini 3.1 Pro [8], with* 5*-way self-consistency—attains* ***81.18%*** *average and* ***78.85%*** *macro-average accuracy on the hidden test split, surpassing the prior leaderboard best* (80.85%)*. The pipeline is inference-only and fully cached for exact reproduction.*

1. Benchmark and Approach Overview

Benchmark. ImplicitQA/VRR-QA [1] evaluates *implicit* reasoning in Video-LMMs: each multiple-choice question is constructed so that its answer cannot be read off any single frame, but must be inferred across frames. The challenge exposes two phases: a **Validation** split of 1,001 QA pairs with *public* labels (our development set) and a hidden **Test** split of 172 QA pairs that decides the final ranking. Questions span nine reasoning categories; roughly 69% are spatial (lateral, vertical, relative depth/proximity, viewpoint). The number of options varies (2–8; the gold-letter prior favors B, so chance $\approx$ 30%). Organizers provide pre-trimmed clips named by `question_id` (mean duration 16.6 s on test). Scoring uses **Average Accuracy** and category-wise **Macro-Average Accuracy**.

Approach. We treat the task as *inference* + *test-time compute*: there is no in-distribution training set, so gains must come from (i) base-model choice, (ii) how video is presented to the model, (iii) prompting, and (iv) test-time aggregation (Fig. 1). We built a modular pipeline and used the labeled validation split to select every design decision before committing it to the hidden test split. The hypothesis that emerged early—and that all ablations confirmed—is that *the bottleneck is perception, not reasoning search*.

2. Architecture, Prompting, and System Design

2.1. Three-layer pipeline

The codebase separates concerns behind stable interfaces (factory pattern + abstract base classes):

- **Data layer:** loads QA pairs, resolves the trimmed clip per `question_id`, uniformly samples frames (cached), and optionally attaches a Whisper audio transcript.
- **Model layer:** a common `predict_batch` interface implemented by (a) a vLLM [7]-backed multimodal model for open weights and (b) a Gemini API model.
- **Runner:** ties data and model via a fixed `QASample`/`Prediction` contract, scores against labels when present, and emits the submission JSON; a data-driven fallback to the majority class ("B") guarantees a valid answer for any failed item.

2.2. Serving

Open models are served with vLLM; frames are passed as multi-image inputs. Because the cluster GPUs are A100-40GB (no NVLink), tensor parallelism deadlocked on NCCL/shared-memory broadcast, so we run **data-parallel** inference (one replica per GPU over a shard, then merge)

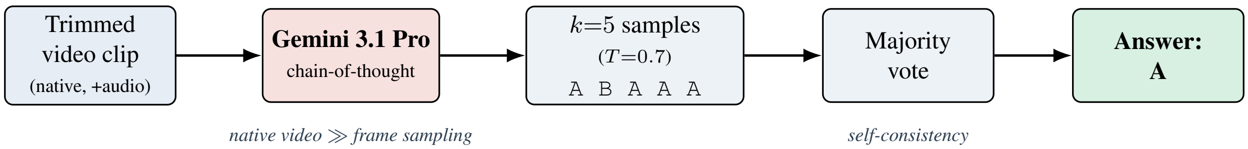

Figure 1. **Overview of the final system.** Each pre-trimmed clip is fed as *native video* to Gemini 3.1 Pro, which produces k=5 independent chain-of-thought answers at temperature 0.7; a majority vote yields the prediction. Native video ingestion supplies the perceptual capability the benchmark demands, and self-consistency denoises the strong base—together reaching 81.2% test accuracy. Uploads and responses are cached, so re-runs are free.

and use 4-bit **AWQ** quantization (the Marlin kernel works on Ampere, whereas FP8 does not) with eager execution to fit large models on a single card. The final system uses **Gemini 3.1 Pro** (`gemini-3.1-pro-preview`), which ingests the *native video* (and audio) rather than sampled frames; each clip is uploaded once and every response is cached to disk.

2.3. Prompting and test-time compute

The default prompt presents the question and (variable-count) options, requests chain-of-thought, and constrains the final line to Answer: <letter>; a robust parser handles <think>/<answer> formats. We also ablate a *reasoner* template, a *spatial* "describe-the-layout-first" template, a two-stage *cascade* (blind description → answer), and a three-stage *decomposition* (atomic sub-questions → answer over frames → aggregate). **Self-consistency** [6] samples k reasoning paths at temperature T and takes the majority vote; this is the one component that transferred from validation to test on a strong base.

3. Training Details and Implementation Setup

Training. *None.* The system is entirely training-free; gains come from base-model selection, video presentation, prompting, and test-time aggregation. **Setup.** Open-model experiments ran on NVIDIA A100-40GB GPUs via SLURM with vLLM 0.18.1 (PyTorch 2.10, Transformers 5.7) under data-parallel sharding; audio uses `faster-whisper`. The frontier system uses `google-genai` 2.7 against `gemini-3.1-pro-preview` and needs only CPU + network. Final configuration: native video input; self-consistency k=5, T=0.7, `max_output_tokens`= 2048. Open baselines used 16 uniform frames at 448px. The full Gemini run cost ≈$11.8; responses and uploads are cached, so re-runs make zero API calls.

4. Results

Table 1 reports validation (n=1001) and test (n=172) accuracy for key systems; Fig. 2 visualizes the test progression. Test is consistently ∼5–8 points *easier* than validation for a fixed system, so we use validation for all selection.

	Val		Test	
System	Avg	Mac	Avg	Mac
Qwen2.5-VL-7B (CoT)	45.8	48.7	50.2	47.3
Qwen3-VL-8B (CoT)	46.7	49.9	54.6	52.3
+ decomposition	46.7	50.3	56.2	53.1
+ route (single↔dec.)	47.5	51.3	56.5	54.8
Qwen3-VL-32B-AWQ (CoT)	48.7	52.5	55.9	56.2
+ self-consistency (k=5)	**51.3**	**54.5**	58.5	56.2
Gemini 3.1 Pro + SC (k=5)	—	—	**81.2**	**78.9**
Prior leaderboard best			*80.9*	—
GPT-o3 [1]	64.1	68.6		
Human (non-expert) [1]	83.0	85.6		

Table 1. Average / Macro accuracy (%). Open-source peaks at 58.5 on test; the frontier native-video model with self-consistency reaches 81.2, surpassing the prior best of 80.9.

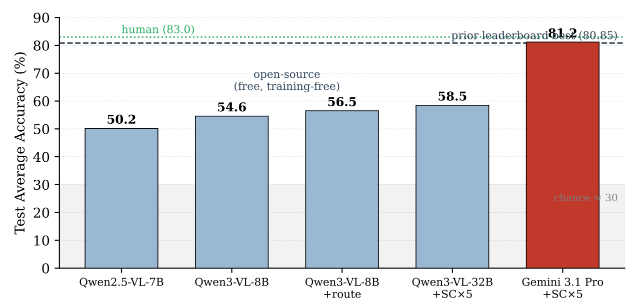

Figure 2. **Test average accuracy across systems.** The training-free open stack climbs from 50.2 to a 58.5 ceiling; the frontier native-video model with self-consistency reaches 81.2, above the prior leaderboard best (dashed) and approaching non-expert human performance (dotted).

5. Ablation Studies and Observations

All ablations are on the labeled validation split (full 1,001 questions unless a 400-question subset is noted). Table 2 summarizes the decisive ones.

Observations.

1. **Self-consistency is base-dependent:** majority voting *denoises* a mostly-correct model but *amplifies* a mostly-

Lever	Setting	Val Avg	Effect	Takeaway
Base scale (CoT, greedy)	Qwen2.5-VL-7B → Qwen3-VL-32B-AWQ	$45.8 \rightarrow 48.7$	$+2.9$	stronger base helps
Self-consistency k=5 on *weak* 7B	T=0.6	42.9	-1.7	voting amplifies a wrong base
Self-consistency k=5 on *strong* 32B	T=0.7	51.3	$+2.6$	voting denoises a right base
Decomposition on *weak* 8B	3-stage	(test $+1.6$)	helps	scaffolding aids weak model
Decomposition on *strong* 32B	3-stage	46.6	-2.1	error propagation hurts strong
Describe-then-reason cascade	blind, 400-sub	29.8	-9.3	blind desc. drops the implicit cue
Spatial "layout-first" prompt	7B	39.7	-6.1	rigid structure hurts
Image resolution	448 → 768px (32B)	50.0	$+1.3$	more pixels help (perception)
Audio transcripts (Whisper)	7B	42.9	-1.5	spatial task, audio is noise
4-way cross-model category routing	test	→ 50–53	overfits	noisy on 172-question test
Native video vs. frame sampling	Gemini 3.1 Pro	large (Tab. 1)		the decisive lever

Table 2. Key ablations on the validation split. Reasoning-side augmentations are neutral-to-harmful; perception-side levers and self-consistency on a *strong* base are the reliable gains.

wrong one—it hurt the 7B (-1.7) yet helped the 32B ($+2.6$) and the frontier model. On the small (172-question) test split, sampled voting is also high-variance (the 32B drew 58.5 at k=5 but 50.0 at k=9).

2. **Decomposition is also base-dependent**—it helps the weak 8B but hurts the strong 32B, which already performs the multi-hop reasoning internally.
3. **Caption/describe-then-reason fails by design:** a question-agnostic description omits the very implicit cue the question targets, starving the answerer—the intended difficulty of "implicit" QA.
4. **Complex routing overfits the tiny test split,** even with a 59%-accurate learned category classifier; only the simpler two-way same-model route transferred.
5. **Perception is the wall:** higher resolution helps and *native video* ingestion is transformational, while audio and reasoning scaffolds do not help on this spatial-dominated benchmark.

6. Summary of Findings

The human–machine gap on ImplicitQA is a **perception/world-model gap**, not a reasoning-search gap; methods that help reasoning-bound tasks (tree search, decomposition, self-critique, state tracking) are neutral-to-harmful here. The two reliable, transferable levers are **(i) a stronger perceptual base**—especially a *native-video* frontier model—and **(ii) light test-time denoising (self-consistency)** on a strong base. Disciplined use of the labeled validation split to reject overfit-prone tricks (cross-routing, naive voting, cascades) was essential: several methods that improved validation *degraded* the 172-question test score. These insights minimized cost: the final run used only single-pass CoT + self-consistency, reaching **81.2%** test accuracy for ≈$12 with full caching for exact reproduction.

Limitations. The 172-question test split makes single-submission scores noisy (± a few points for stochastic decoding); our margin over the prior best is narrow and is being reinforced with higher-k self-consistency and a second frontier voter. "Reasoning > scale" comparisons are on this benchmark only.

References

[1] S. Swetha, R. Gupta, P. Kulkarni, et al. ImplicitQA / VRR-QA: Visual Relational Reasoning in Videos Beyond Explicit Cues. *arXiv:2506.21742*, 2026.

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[6] X. Wang, et al. Self-Consistency Improves Chain-of-Thought Reasoning in Language Models. *ICLR*, 2023.

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